

Programming in C

Lab 8: **struct** and **union**

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1 struct

2 union

struct (1)

- Define a **struct** for Complex number (**Comp**x)
 - real (**real**) and virtual (**virt**) part
- Define a function to perform multiplication between two complex numbers
- **struct** Comp_x multComp(**struct** Comp_x d1, **struct** Comp_x d2)
 -
 - $rslt.rl = a.rl * b.rl - a.vt * b.vt;$
 - $rslt.vt = a.rl * b.vt + b.rl * a.vt;$
 - ...
- Define a function “**void printComp(struct Comp a)**” to print out a complex number
 - It prints like following:
0.3+0.5i

struct (2): the answer

```
1 #include <stdio.h>
2 struct Comp{
3     float rl;
4     float vt;
5 };
6 struct Comp multComp(struct Comp d1, struct Comp d2)
7 {
8     struct Comp r;
9     r.rl = d1.rl*d2.rl-d1.vt*d2.vt;
10    r.vt = d1.rl*d2.vt+d2.vt*d1.rl;
11    return r;
12 }
13 void printComp(struct Comp r)
14 {
15     if(r.vt > 0)
16         printf("%f+%fi\n", r.rl, r.vt);
17     else if(r.vt < 0)
18         printf("%f%fi\n", r.rl, r.vt);
19     else
20         printf("%f\n", r.rl);
21 }
```

struct (3): the answer

```
22 int main()  
23 {  
24     struct Compz d1 = {1.2, 5.3}, d2 = {1.2, -1.3};  
25     struct Compz r = multComp(d1, d2);  
26     printComp(r);  
27     return 0;  
28 }
```

struct array (1)

- Define a **struct** named **NoteBook**
 - qq: number, **int** type
 - name[32]: friend's name, **char** type
 - phone[16]: phone number, **char** type
- Define an array (**5 elements**) of NoteBook
 - Input five records
 - Output five records
- Please check the size of your defined **struct** type

struct array (2)

```
1 #include <stdio.h>
2 struct Notebook {
3     long qq;
4     char name[32];
5     char phone[16];
6 };
7 typedef struct
8     Notebook QQBook;
9 void printQQbook();
10 int main()
11 {
12     printQQbook();
13     return 0;
14 }
```

```
1 void printQQbook() {
2 {
3     QQBook persons[4];
4     int i = 0;
5     for(i = 0; i < 4; i++)
6     {
7         printf("Name: -");
8         scanf("%s", persons[i].name);
9         printf("QQ: -");
10        scanf("%d", &persons[i].qq);
11    }
12    for(i = 0; i < 4; i++){
13    {
14        printf("Name: -%s\n", persons[i].
15            name);
16        printf("QQ: -%d\n", persons[i].qq);
17    }
18 }
```

Outline

1 struct

2 union

union (1)

- Define a **union** type
 - One **float** number
 - One **short** number
 - One **char** character
- Use **typedef** to define type '**DATA**' of above union type
- Declare variable d1 of type **DATA**

union (2)

```
1 #include <stdio.h>
2 union Data {
3     float f;
4     char c;
5     short i;
6 };
7 typedef union Data DATA;
8 void testUnion();
9 int main()
10 {
11     testUnion();
12     return 0;
13 }
```

```
1 void testUnion()
2 {
3     DATA d1;
4     printf("Size of data: %d\n",
5           sizeof(DATA));
6     d1.c = 'a';
7     printf("%c\n", d1.c);
8     d1.f = 3.1415;
9     printf("f: %f\n", d1.f);
10    printf("d: %d\n", d1.i);
11    printf("c: %c\n", d1.c);
12    d1.i = 9;
13    printf("f: %f\n", d1.f);
14    printf("i: %d\n", d1.i);
15    printf("c: %c\n", d1.c);
16 }
```